

Linearity of Expectations

- A function f is linear if $f(ax) = af(x)$ and $f(x+y) = f(x) + f(y)$
- If $y = ax + b$, then the function is affine, but commonly termed as linear
- Expectation is not a function but an operator — it takes a function as input — and it is linear in the same sense.

$$\begin{aligned}
 E(a g(X) + b) &= \int_{-\infty}^{+\infty} (a g(x) + b) f_X(x) dx \\
 &= \int_{-\infty}^{+\infty} a g(x) f_X(x) dx + \int_{-\infty}^{+\infty} b f_X(x) dx \\
 &= a E(g(X)) + b
 \end{aligned}$$

Must know

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Variance

- Shifting a random variable does not change its variance
- Scaling it scales the variance by the square.

$$\begin{aligned}
 \text{Var}(aX + b) &= E[(aX + b - E(aX + b))^2] = E[(aX + b - (a\mu + b))^2] \\
 &= E[a^2(X - \mu)^2] = a^2 E[(X - \mu)^2] = a^2 \text{Var}(X)
 \end{aligned}$$

Must know

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Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three (and that's it)
 - Male and female children are equally likely and independent.
 - Let B (G) denote random variables for boys (girls)
 - What is the $P(B = 0)$?

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Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three (and that's it)
 - Male and female children are equally likely and independent.
 - Let B (G) denote random variables for boys (girls)
 - What is the joint pmf $p(B, G)$
- No children? $P(B = 0, G = 0) = 0.15$
- One girl, no boy? $P(B = 0, G = 1) = 0.2 \times 0.5 = 0.1$

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Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three. What is the joint pmf $p(B, G)$
- Two boys and one girl? $P(B = 2, G = 1) = 0.3 \times (1/8) \times 3 = 0.1125$
 - All 8 combinations of three children are equally likely; three of them have two boys and one girl.

$i \backslash j$	0	1	2	$i \backslash j$	0	1	2	3	Row Sum $= P\{B = i\}$
0	.15	.10	.0875	0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	2	.0875	.1125	0	0	.2000
3	.0375	0	0	3	.0375	0	0	0	.0375
Column Sum $= P\{G = j\}$					.3750	.3875	.2000	.0375	

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Continuous Random Variables

- Recall: X is a continuous random variable if there exists a non-negative function $f(x)$ defined for all real x having the property that for any set C of real numbers

$$\Pr\{X \in C\} = \int_{x \in C} f(x) dx$$

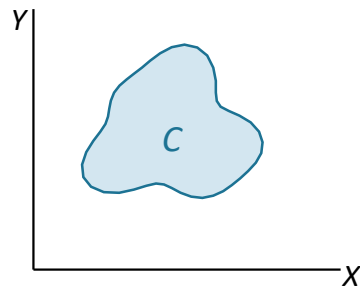
- We say that X and Y are jointly continuous if there exists $g(x, y)$ having the property that for any set C of pairs of real numbers

$$P\{(X, Y) \in C\} = \iint_{(x, y) \in C} g_{X, Y}(x, y) dx dy$$

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Continuous Random Variables

- The function $g(x, y)$ is called the joint probability density function (pdf)
- The joint probability that (X, Y) falls in any region of the plane is the integral of the joint pdf over that region.



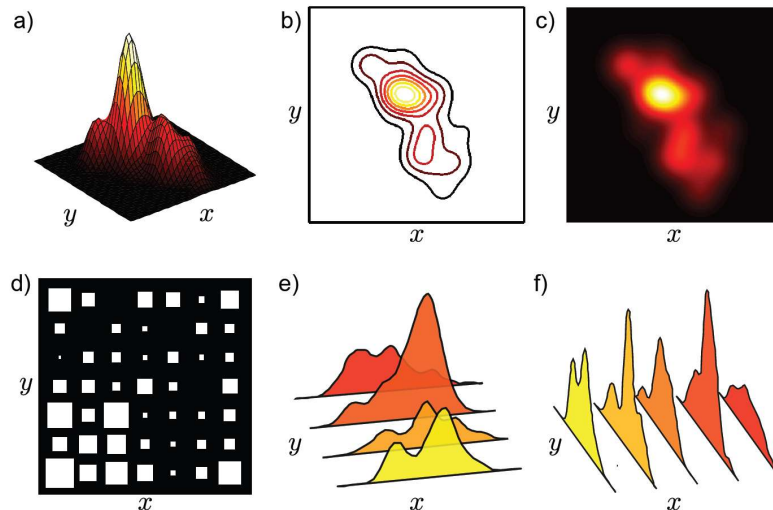
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Example

- A student has four hours of free time in the evening.
- X = hours spent gaming, Y = hours spent streaming videos
- The feasible values satisfy $X \geq 0$, $Y \geq 0$, $X + Y \leq 4$
- Suppose all feasible pairs are equally likely
 - $f_{X,Y}(x, y) = \begin{cases} \frac{1}{8}, & x \geq 0, y \geq 0, x + y \leq 4 \end{cases}$
 - $f_X(x) = \int_0^{4-x} f_{X,Y}(x, y) dy = \int_0^{4-x} \frac{1}{8} dy = \frac{4-x}{8}, \quad 0 \leq x \leq 4$
 - Marginalization adds the probability density along a vertical slice of the joint-density region

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Visualizing Joint Probability



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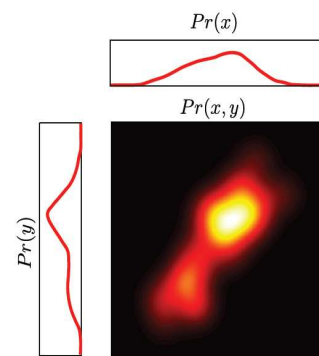
Marginalization

- We can recover probability distribution of any variable in a joint distribution by integrating (or summing) over the other variables
- The marginal pdf of one variable is obtained by integrating the joint pdf over the other:

$$Pr(x) = \int Pr(x, y) dy$$

$$Pr(y) = \int Pr(x, y) dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

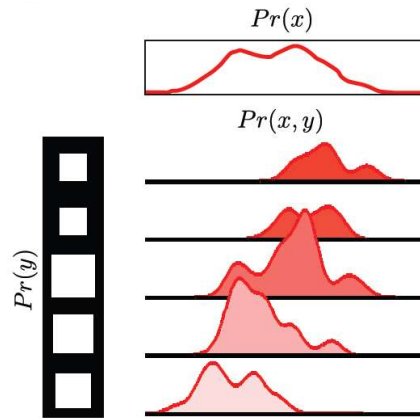


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Marginalization

$$Pr(x) = \sum_y Pr(x, y)$$

$$Pr(y) = \int Pr(x, y) dx$$



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Marginalization

- Works in higher dimensions as well
 - leaves joint distribution between whatever variables are left

$$Pr(x, y) = \sum_w \int Pr(w, x, y, z) dz$$

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Expected Value

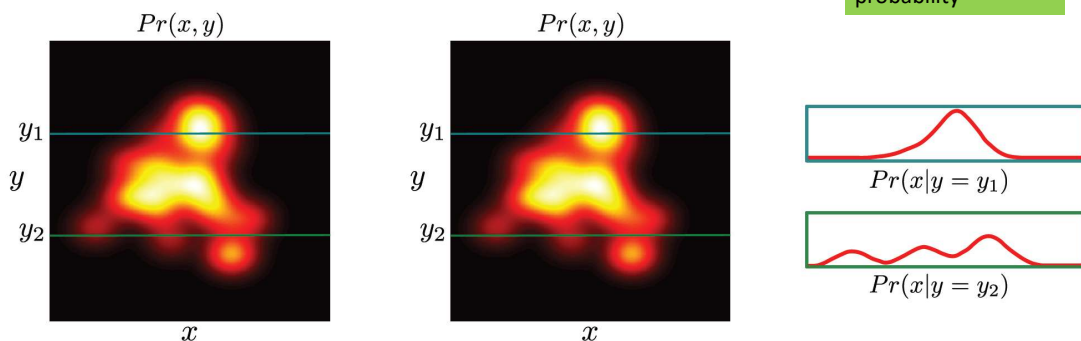
- Expected values of sums of random variable
 - $E[X] + E[Y] = E[X+Y]$
- Example: Expected value of the sums of two (fair) dice
 - Option 1: pmf: $(1/36)[1\ 2\ 3\ 4\ 5\ 6\ 5\ 4\ 3\ 2\ 1]$
 - Expectations: $2*1/36 + 3*2/36 + \dots + 12(1/36) = 252/36 = 7$
 - Option 2: Name the dice as X_1 and X_2
- This is a very useful result, particularly because it makes no assumptions on X and Y
- This is a consequence of the general result

$$E(g(X, Y)) = \int \int_{-\infty}^{+\infty} g(x, y) f_{X,Y}(x, y) dy dx$$

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Slice of the joint pdf

- Earlier we take the sum of the probabilities to recover the marginal distribution
- What happens if we take a slice of the joint pdf as is?



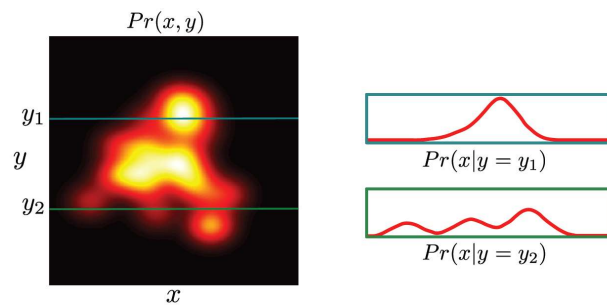
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Conditional Probability

- Conditional probability can be extracted from joint probability
- Extract appropriate slice and also normalize

$$Pr(x|y = y^*) = \frac{Pr(x, y = y^*)}{\int Pr(x, y = y^*)dx} = \frac{Pr(x, y = y^*)}{Pr(y = y^*)}$$



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Conditional Probability

- More usually written in compact form

$$Pr(x|y) = \frac{Pr(x, y)}{Pr(y)}$$

- Can be re-arranged

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$Pr(x, y) = Pr(y|x)Pr(x)$$

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Conditional Probability

- This idea can be extended to more than two variables

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$\begin{aligned} Pr(w, x, y, z) &= Pr(w, x, y|z)Pr(z) \\ &= Pr(w, x|y, z)Pr(y|z)Pr(z) \\ &= Pr(w|x, y, z)Pr(x|y, z)Pr(y|z)Pr(z) \end{aligned}$$

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Reorganizing

$$Pr(x, y) = Pr(x|y)Pr(y)$$

$$Pr(x, y) = Pr(y|x)Pr(x)$$

Combining:

$$Pr(y|x)Pr(x) = Pr(x|y)Pr(y)$$

Re-arranging:

$$\begin{aligned} Pr(y|x) &= \frac{Pr(x|y)Pr(y)}{Pr(x)} \\ &= \frac{Pr(x|y)Pr(y)}{\int Pr(x, y) dy} \\ &= \frac{Pr(x|y)Pr(y)}{\int Pr(x|y)Pr(y) dy} \end{aligned}$$

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Bayes' Rule Terminology

Likelihood – propensity for observing a certain value of x given a certain value of y

Prior – what we know about y before seeing x

$$Pr(y|x) = \frac{Pr(x|y)Pr(y)}{\int Pr(x|y)Pr(y) dy}$$

Posterior – what we know about y after seeing x

Evidence – a constant to ensure that the left hand side is a valid distribution

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